Shahjalal University of Science and Technology
Department of Computer Science and Engineering
4th Year 1st Semester Final Examination' Jun 2023 (Session: 2019-20)
Course Code: CSE 421 Credits: 2 Course Title: Software Project Management
Time: 2 hrs Total Marks: 50

Group A

[Answer all the questions]

1. Write any FIVE **5x1=5**

a) What are the four P's that effective software project management focuses on?
b) Define Feasibility Study.
c) What is a contract?
d) What is the cost-benefit analysis?
e) What is meant by cost flow forecasting?
f) What is a Critical path?
g) What parameters are involved in computing the total cost of a software development project?
h) What is Masslow's hierarchy needs?

2. Answer any FOUR **4x2.5=10**

a) What are the characteristics of software projects that make them more difficult than other projects?
b) What is project portfolio management? Explain the key aspects of project portfolio management.
c) What is the use of decision tree in risk evaluation?
d) How would you select the right person for a job?
e) Explain the drawback of the waterfall model. Also, discuss the spiral model with its merits and demerits.
f) How to manage activity risks?

3. Answer any TWO **2x5=10**

a) What are the activities covered by software project management? Explain with a diagram.
b) Construct a precedence diagram to analyze your project. Show the critical path on your precedence diagram.

Activity	Duration (week)	Predecessors

on?

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c) What is a contract?

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Activity	Duration (week)	Predecessors
A	1	-
B	2	-
C	2	A
D	4	A
E	1	B
F	2	C, D
G	3	E
H	1	G
I	4	G
J	1	F
K	3	J, H
L	4	I
M	1	K, L

c) Briefly describe the stages of a contract placement

Group B

[Answer all the questions]

4. Write short notes (Any FIVE) **5x1=5**

a) What is the simplest model of software development paradigm?
b) What is PRINCE2?
c) Define project stakeholders.
d) To detect fraudulent usage of credit cards, what data mining task should be used?
e) What is a bespoke system?
f) What is requirement Analysis?
g) What is Work Breakdown structure?
h) What is the payback period?

5. Answer any FOUR **4x2.5=10**

a) What do you understand by Egoless Programming?
b) Explain the Oldhun-Hackman job characteristic model.
c) What are the objectives of activity planning?
d) What are the factors that made the job meaningful?
e) What are the advantages & disadvantages of fixed price contracts?
f) Define risk assessment using formula.

6. Answer any TWO **2x5=10**

a) Discuss leadership models. Explain function of a leader with an example
b) Briefly explain about Stepwise project planning.
c) Based on the data given below. Calculate Net Profit, Payback, and ROI for each project

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000

Group B

[Answer all the questions]

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3	20,000	200,000	30,000	30,000
4	20,000	200,000	20,000	25,000
5	100,000	350,000	20,000	50,000